An on Board Charger and Wireless Receiver Integrated Topology Based on Decoupled Magnetic Circuit and Multifunction Power Bridge

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***Abstract*—To further improve the system integration and reduce the system cost of the batter charging system for electric vehicles, this article proposed a charging solution that integrates bidirectional on-board charger (OBC) and wireless power transfer (WPT) based on decoupled magnetic circuit and multifunction power bridge. The proposed solution presents multiple highlights: 1) It only uses one active bridge in the HV side because the bridge is totally reused by OBC and WPT. 2) WPT mode is almost totally decoupled with OBC mode and not add extra switch so it has lower cost and higher efficiency. This article includes the description of the system topology, operation mode of OBC and WPT, and the design method of system parameters. Finally, a prototype was constructed and different charging modes were tested to verify the proposed topology.**

Keywords*—wireless power transfer (WPT), on-board charger (OBC), electric vehicles (EVs), integration and decouple, non-extra mechanical switch

I. INTRODUCTION

Wireless power transfer (WPT) is one of the most advanced charging technology. It is based on the coupled-coils to transfer power, so the TX-coil and RX-coil have not electrical connection. Because of its high-efficiency, convenience, and the popularity of autonomous driving technology, WPT is widely adopted in electric vehicles (EVs) as charging systems [1] [2].

As the popularity of autonomous driving technology, EVs which have automated parking system (APK) also have the need of automated charging system (ACS). Due to the convenience and safety of WPT, it quickly became one of the most competitive ACS, but it is usually expensive with limited charging power. The on-board chargers (OBC) usually have higher charging speed but have less charging convenience than the WPT. EVs equipped with both on-board charger (OBC) and WPT allow users to select, but it could be costly and huge.

Integrating the OBC and WPT mechanically can make the system more compact, but it does not reduce the system cost. Charging systems in the future are more expected to focus on magnetic or topological integration of WPT and OBC. Reference [3] proposes an integrated topology of the WPT and OBC systems based on sharing an active bridge. However, the OBC and WPT modes are not decoupled in this system. In the OBC mode, current will be generated in the RX coil which may cause EMI; while in the WPT mode, it will charge the OBC primary-side capacitor, which may cause to the system

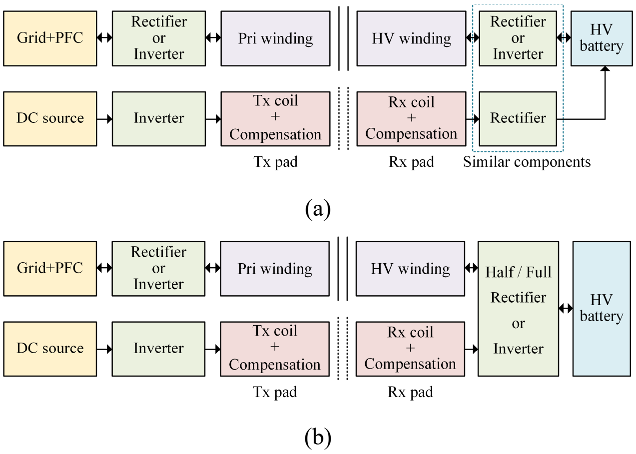

Fig.1. (a) Block diagram of the conventional WPT and OBC integration; (b) Block diagram of the proposed integrated topology based on decoupled magnetic circuit and multifunction power bridge.

get damaged. Reference [4] makes the OBC transformer similar to the WPT coupled-coils and uses a mechanical switch in the primary resonant tank, so it has no need for an additional RX-coil to achieve the function of WPT. However, not only current will be still generated in the WPT-coil during the OBC mode and cause EMI, but also the mechanical switch will reduce efficiency of the system. Reference [5] is similar with Reference [4], saving the space of the OBC transformer, but still cause EMI by the WPT coil during the OBC mode. Therefore, the integration of WPT and OBC not only requires the decouple of them during independent operation, but also need to consider both efficiency and system cost.

To achieve the integration and the decouple of WPT and OBC, an integrated topology of WPT and OBC based on decoupled magnetic circuit and multifunction power bridge is proposed in this article. Fig.1 (a) and (b) respectively show the block diagram of the conventional mechanically integration solution and the proposed integrated solution. Compared with the mechanically integration and the previous integration in articles [3] [4] [5], the proposed integrated topology has the following benefits: 1) It has only one active bridge in the HV side because the bridge is totally reused by OBC and WPT. 2) WPT mode is almost totally decoupled with OBC mode, and this solution has no need of extra switch so that it will have less EMI and higher efficiency.

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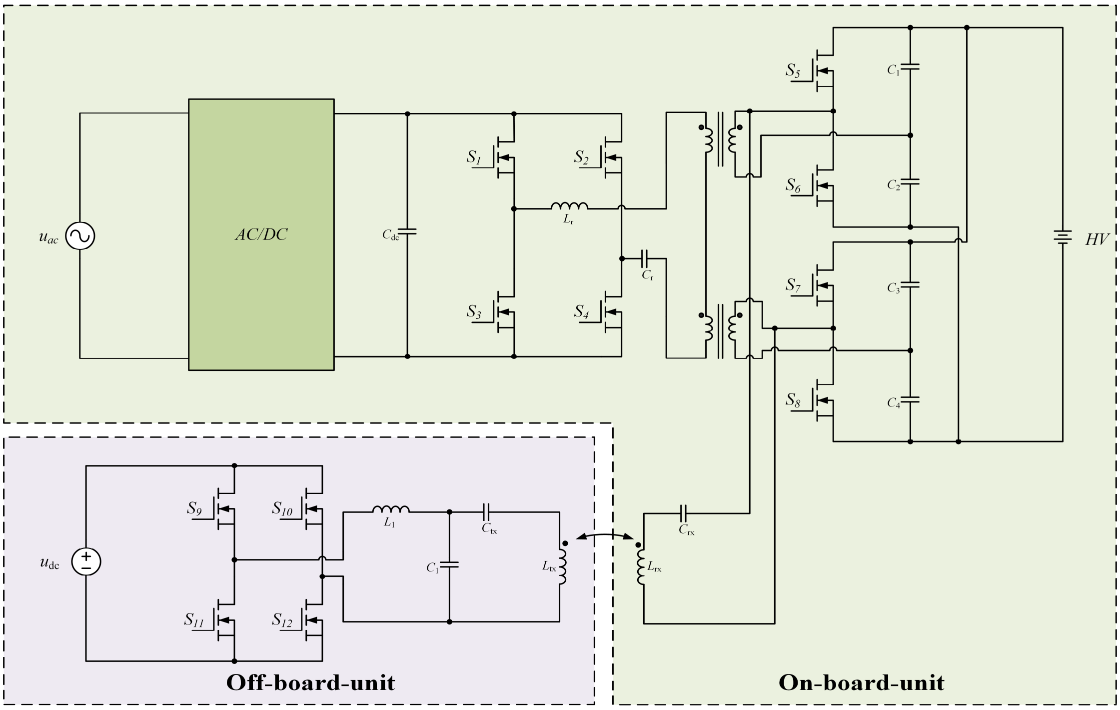

Fig.2. The proposed integrated EV charger's topology.

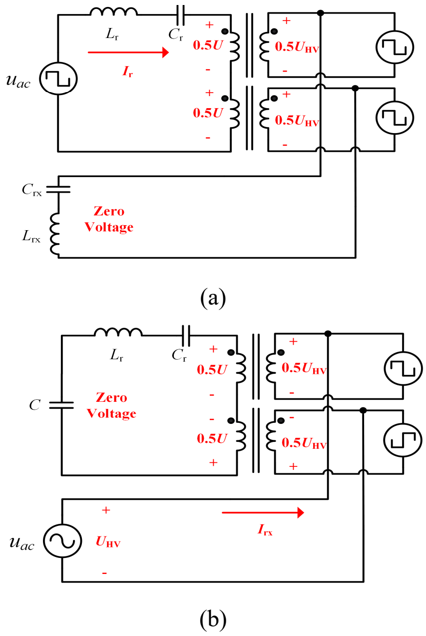

Fig.3. The diagram of how the integrated topology makes WPT decoupled with OBC. (a) OBC operation mode. (b) WPT operation mode.

II. Integration Method

A. Topology

The proposed integrated EV charger's topology is shown in Fig.2. In the primary side of OBC, a high frequency inverter is composed by MOSFET S_1-S_4. The primary side of two transformers are connected in series, the secondary sides of two transformers are respectively connected with two voltage-doubler rectifiers which are composed with S_5-S_8 and C_1-C_4. Then two rectifiers are parallelled and connected with HV. The resonant tank is composed with L_r and C_r so the primary side and secondary side of OBC constitute the series-resonant-dual-active-bridge-converter (DBSRC). The LCC-S compensation which consists of L_1, C_1, C_{tx}, C_{rx}, is adopted for WPT. Rx-coil is connected with the secondary-sides' dotted terminals of two transformers then the two half bridges are reused as an active bridge rectifier for WPT, and the half bridge capacitors are reused as dc-link capacitors for WPT.

B. Operation Mode

When the system works in the OBC mode, it uses pulse-frequency-modulation and phase-shift-control in the same time to control the output power. During this mode, switch S_5 and S_7 are controlled by the same pulse, S_6 and S_8 are controlled by the other complementary pulse. Then the voltages of two terminals of L_{rx} and C_{rx} always keep same like Fig.3 (a), either HV voltage or zero voltage, so that it will hardly cause current in the Rx-coil. The switch frequency of OBC is around 200kHz and is much higher than the frequency of WPT (85kHz) then the WPT Rx resonant tank will have a larger impedance at 200kHz, so that it can reduce the current in the Rx-coil which is caused by the non-ideal factor such as the delay of drive pulses to the MOSFETs' gates so that it will make less EMI than the previous system.

In the WPT mode, the two half bridges are reused as an active bridge. It uses current-sensor to achieve the zero-current-detection-based synchronized rectifier control at the Rx side to improve efficiency [6] [7]. The secondary sides' voltages of two transformers is like Fig.4 (b) which are equal but opposite. Then the voltages of primary sides are also equal but opposite. As the primary sides of transformers are connected in series, the voltage between the OBC primary bridge's AC-side is almost zero, then the primary side's capacitor will not be charged to high voltage. This method makes WPT decoupled with OBC and promises the safety of the system.

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III. Parameter Design

OBC topology is DBSRC, the resonant frequency is designed at 100KHz, and the switch frequency is designed at between 150kHz and 250kHz, so that it has two variables to modulate the output power.

The resonant tank is the most important parameter for the converter to design. The output power follows the equations:

$$F = f_s / f_r \tag{1}$$

$$Q = \omega_r L / Z_B \tag{2}$$

$$M = nU_o / U_{in} \tag{3}$$

TABLE I. Parameters of the On-board Unit

L_r	C_r	M (n_1:n_2)	L_{rx}	C_{rx}
45μH	56.4nF	24:19	91μH	39.6nF

TABLE II. Parameters of the Off-board Unit

L_1	C_1	C_{tx}	L_{tx}	k
40μH	99nF	66nF	91uH	0.32

$$P_{omax} = \frac{8M \sin\phi}{\pi^2 Q(F-1/F)} \tag{4}$$

The doubler rectifiers' capacitor is reused as DC-link capacity and dc blocking cap, so it must have larger capacity than usual.

WPT switch frequency is around 85KHz, its parameter design is based on LCC-S compensation, L_1 almost decides the maximum output power [8]:

$$L_1 = \sqrt{\frac{k_{max} U_{in} U_{HV} L_{tx}}{\omega_0 P_{max}}} \tag{5}$$

The value of C_1 can be calculated from (6) as follows:

$$C_1 = \frac{1}{\omega_0^2 L_1} \tag{6}$$

C_{tx} can be calculated from (7) as follows:

$$C_{tx} = \frac{1}{\omega_0^2 (L_{tx} - L_1)} \tag{7}$$

C_{rx} can also be calculated from (8) as follows:

$$C_{rx} = \frac{1}{\omega_0^2 L_{rx}} \tag{8}$$

In the end, tuning L_1 to achieve the zero-voltage-switching (ZVS) operation conditions for improving the efficiency.

IV. Experimental Verification

As shown in Fig.4, a prototype including the on-board unit and off-board unit was constructed. The topology is shown in Fig.3. The input voltage is 400V and the output power is 1.5kW. The parameters of the on-board unit are listed in Table I, the SIC MOSFET was used in this converter, the switching frequency was around 200kHz and the dead time for each bridge was 200ns. The parameters of the off-board unit are listed in Table II, the switching frequency is around 85kHz and in this condition is 80kHz. A resistive load of nearly 50Ω is used for the OBC and WPT mode.

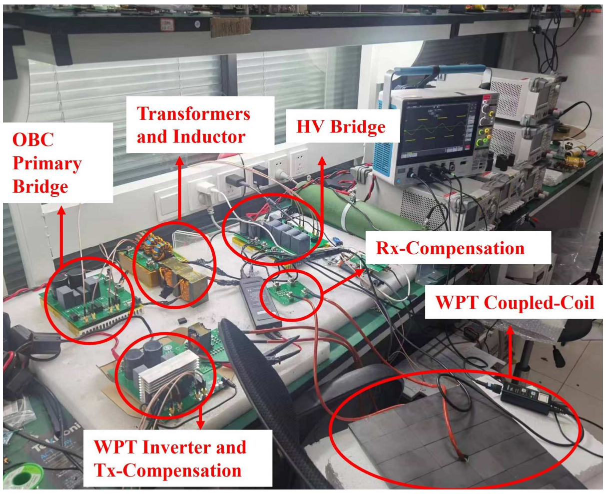

Fig.4. The prototype including on-board unit and off-board unit.

The key waveform of OBC and WPT operation mode is shown in Fig.5 and Fig.6. It can be seen that the system can finish the function of WPT and OBC and perfectly achieve ZVS. The peak efficiency of OBC operation mode is 97% and WPT mode is 92% at 1.5kW.

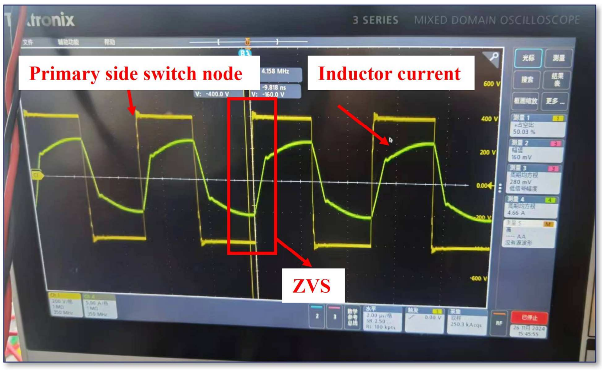

Fig.5. The prototype including on-board unit and off-board unit.

The experiment also tests the result of decoupled design. In the OBC mode, the voltage between the Rx-coil is almost zero, and in WPT mode, the voltage between the OBC primary side's capacitor is always below 1V during the experiment. It proves that WPT is almost totally decoupled with OBC based on the decoupled magnetic circuits.

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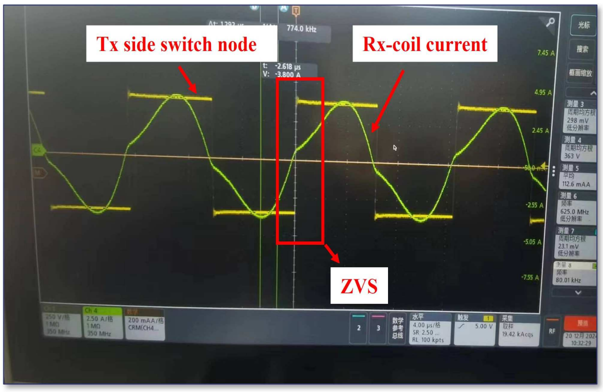

Fig.6. The prototype including on-board unit and off-board unit.

V. Conclusion

This article proposes an on-board charger and wireless receiver integrated topology based on magnetic decoupled circuit and multifunction power bridge, offering WPT and OBC functionalities within a single unit. By reusing the OBC half-bridge rectifiers as a full bridge rectifier for WPT, the topology reduces the cost of an active bridge compare with the standalone WPT and OBC method. By designing the connection of two transformers and make a decoupled magnetic circuit, the WPT is totally decoupled with the OBC so that when the OBC working it will not make current at the WPT coil, and when the WPT working it will not charge the OBC primary side's capacitor. This results in a volume reduction and corresponding cost decreases. Experimental results affirming the efficacy of this integrated topology.

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